

Brain Age Prediction

Chronological age regressed from 832 anonymised MRI morphometry features - deliberately corrupted with missing values, outliers and irrelevant columns.

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0.7713

best public leaderboard R^2

1212 × 832

train rows × features, 776 test rows

29

logged experiment rounds

Experiments

STEP	R ²	Δ
Baseline, raw features	0.4584	—
+ median impute, robust scale	0.4414	−1.72%
+ outlier removal, IForest 0.04	0.4542	+1.28%
+ feature selection, k = 125	0.4994	+4.52%
+ best single model, SVR	0.5288	+2.93%

LOCAL CV 0.5807 → PUBLIC LB 0.7047 +12.4 %

PUBLIC LEADERBOARD R², BARS FROM A 0.68 BASELINE

0.7047

5-base stack + 3% label-noise removal

0.7383 +3.4%

+ TabPFN as a decorrelated 6th base

0.7579 +2.0%

+ MICE-ET imputation

0.7713 +1.3%

The Final Space

Imputation	Outliers	Scaling	Features	Label noise	Model
MICE-ET	Winsorize 0.5 / 99.5	Quantile, normal output	SelectKBest, k = 125	SVR OOF, worst 3%	6-base stack, LassoCV meta
Median					
KNN, k = 5 / 10 / 20	No treatment	Robust scale	k sweep, 40 → 400	Keep every row	Tuned singles, 6 families
MICE-Bayesian	IForest 0.04 / 0.10 / 0.20	Power transform	Mutual info, SFM, RFE	Worst 5% / 8% / 10%	CatBoost, GPR, ARD, MLP
Ridge		n_quantiles 100 / 300 / 800			
Transductive MICE	LOF-20 / LOF-35	Target transforms	Ratios and composites	Stack-mean flagging	2nd TabPFN on PCA
	LOF on a PCA view		L/R asymmetry, log, residualised	IForest rescue	Hill-climb, NNLS, equal blend

Above each rule: what shipped. Below it: what was measured and dropped. Every axis was moved alone, against fixed folds, so each delta is a paired comparison.

Rejected Experiments

Feature engineering, five families

–0.013 to +0.001 over 3 seeds — inside a 0.02 seed spread, even after MICE cleaning.

A 7th base learner — MLP, PCA-TabPFN

Standalone 0.585 and 0.536, both ≥ 0.89 correlated — the meta down-weights them to nothing.

Hand-tuned blends over the meta

Hill-climb 0.695, NNLS 0.698 vs the LassoCV meta at 0.702.

Also rejected: CatBoost, GPR, ARD, target transforms, robust GBM objectives, missingness flags, $k > 150$. The SVR + TabPFN core saturates the stack — only a member both strong and decorrelated ever moved it.

Stack-mean 5% noise flagging

Proxy said +0.018; the board said –0.034. It was dropping hard subjects, not bad labels.

Two-signal noise detector

Rescuing 19 feature-anomalous rows from the drop set cost 0.016.

Transductive MICE on train + test

+0.001, and the test set's 10^8 columns poison the round-robin.
